



New in this release 1.9-A - 07/06/2026

"Robin Hood and his Merry Men."

This update to WaveSteward includes several major changes to its capabilities. And that's thanks to an archer who gave me some ideas.

Added

1. A generic spam functionality has been added, which is character-bound. More information below in the document.

Generic Spam {Any glyph, char-bound}		Save Settings
Skill Identifier:	0	... skill
Glyph:	glyph	
Cancel Delay:	120	milliseconds
Recast Delay:	10	milliseconds

Modified

1. **[Bug]** -- There was a slight logic bug in the Lifehunter spam, as also in the Pangu Lifehunter spam, and the Cleric Cyclone Spam. The Cancel and Recast delays were not correctly used. This has now been updated and should be working more correctly
2. **[Update]** -- The standard spams, using the hardcoded delays, have been moved from 100/10 to 120/10. If other values are wished, please use the User-Defined spams you can configure in Misc >> Skill

Settings.

3. **[Update]** -- The User Defined spams have been revised, and a minor protection has been built in that no erroneous values can be inserted.
4. **[Update]** -- The Notification window that pops up at the bottom right of your screen now includes in its title the name of the character the notification is related to.
5. **[Bug]** -- The Multi Lifehunter spamming ability was reported to fail. The code was verified, and some minor quirks were removed. As far as it has been tested for now, it seems to work correctly. Please feel free to test and report any issues you notice.

A very important remark -- To keep in mind. :

Due to changes in WaveSteward's Daily tab, please verify your settings there, as some of your dailies may have shifted. I tried to avoid it as much as possible.

Generic Spam

So, starting with this release, a more generic spam ability has been added.

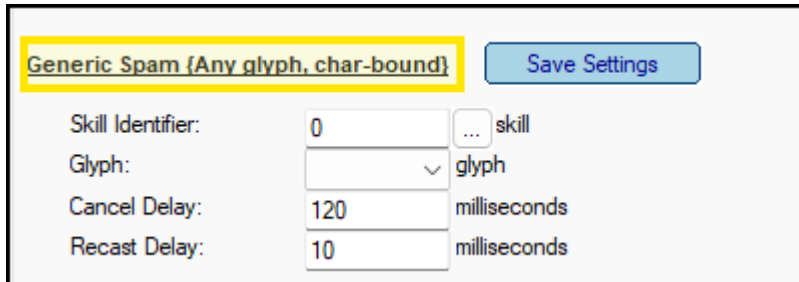
This means that from now on, WaveSteward possesses the ability to spam with any skill that has been glyph'd, with the stress on glyph'd.

But this will also mean some additional configuration to use it if you want. In the following steps, the configuration will be explained.

And as a second note, this is experimental for the users. We don't know the perfect delay settings for all the different characters, so it will be up to you to find the best delay settings when you want to use it and share them with each other, or send them over so we can investigate how to include them.

Configuration

The configuration of the General Spam is on the Misc >> Skill Settings tab.



The screenshot shows a configuration window titled "Generic Spam {Any glyph, char-bound}" with a yellow highlight. To the right of the title is a blue "Save Settings" button. Below the title, there are four rows of settings:

Skill Identifier:	0	... skill
Glyph:	glyph	glyph
Cancel Delay:	120	milliseconds
Recast Delay:	10	milliseconds

You need to do this configuration with the client running for the character you want to configure, and have that character selected in WaveSteward.

The settings are character-bound!!!

One of the main problems in PWI is that, for the system, skills are numbers, and translating these numbers is not that easy or straightforward. For this reason, a workaround was implemented in WaveSteward for quickly detecting the skill we want to use.

Detecting the skill identifier you want to use

Unless you know the number of the base skill you want to use, which probably isn't the case, you will have to detect it. And that can be easily done because of a funny way of storing the glyph'd skills in memory.

1. In PWI, open your Skill List window by pressing R



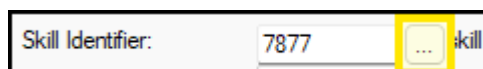
2. Go to the Glyphs



3. Modify the glyph of the skill you want to use. The suggestion is, for example, to remove the assigned glyph and set it back.

4. Now go to WaveSteward, to the tab Misc >> Skill Settings

5. You will see a small button next to the Skill Identifier field with 3 dots.



6. Click on that button.

7. Et voila, the skill identifier field will be filled in with the identifier for the skill, and below you will see the colour of the glyph assigned to it.

Easy peasy lemon squeezy.

8. Now you can also set a Cancel and Recast Delay, then click the Save Settings button to store it for your character.

9. You can also assign a Hotkey to the Generic Spam, which you can find in the list on the Hotkeys tab.

Sin LH Spam 120+10	
Sin LH spam User	lx
PANGU LH Spam 120+10	
PANGU LH Spam User	^x
Multi LH Spam	!^x
Generic Spam	^ls
Cleric Cyclone 120+10	
Cleric Cyclone User	

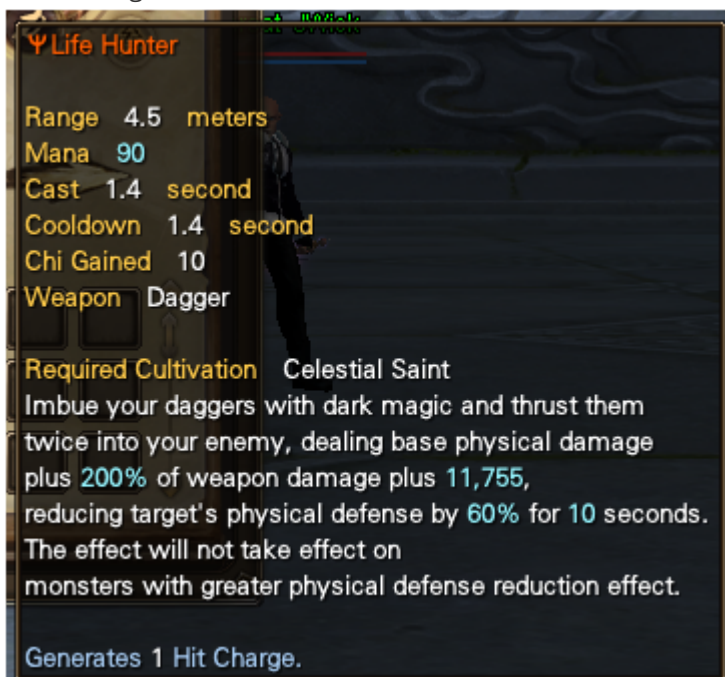
VERY Important!!!

⚠ Warning

Not all skills are good for spamming. That is known.

The main restrictions are:

1. Use of Chi - If a skill needs Chi to be cast, you know that skill is not spammable. Don't even try it.
2. Channelling - If a skill has a channelling setting, this will also affect the functionality. You will have to experiment with the Cancel Delay to be sure that the skill reaches its target. But this also means that if you work on your equipment to reduce the channelling time, it will help you spam faster.
3. Distance - Spamming needs to happen from a very short distance of your target (or better said, from the target's hit box).
4. Cast and Cooldown - The values for a skill for Cast and Cooldown also limit the spam ability of the skill. If these values are high, it means no spamming can occur. For example, the values for an Assassin and Lifehunter are around 1.4 seconds, which is decently high, but the skill has no channelling.



So feel free to try it out and experiment with the values. And find out if it can be of any help to you while in PWI.

Things in the pipeline

1. Solution for long-standing issues:
 1. [Speed Dailies] is going in a loop and stopping to react
 2. [Auto-banking] is not performing
 3. Failure to enter the homestead because it is not in a safe zone.
2. Ability to interrupt WaveSteward without requiring the user to close the application.
3. Extend WaveSteward so that Teleport Stones are no longer an obligation to move through and between the maps.
4. Possible material dig spamming.

Dreams

1. Ability to predict the complete set of bosses and mobs in the WM instance. (We are getting there slowly...)
2. Continue revamping the historically overloaded screens, which is still ongoing.
3. A 64-bit WaveSteward...

Although I know a couple of people are already sweating on it, I don't want to add my sweat to it. And secondly, 64-bit seems rather laggy and more rollercoaster-y due to rubberbanding.

But OK, still dreams, so don't ask when...

Important Information

Please be aware of the following points when using the WaveSteward.

1. WaveSteward is designed to be used with a 32-bit client of PerfectWorld International. Although PWI no longer supports 32-bit, and the default installation from scratch no longer includes the application as 32-bit, it is still possible to extend the installation with a 32-bit client.
2. When running 32-bit, please make sure the following important configurations are in place
 1. Quest list is not displayed on the HUD when you are playing - Origin of crashes in the Eastern Continent and Tremelo
 2. You Graphics quality is set to the minimum for all, with the exceptions:
 1. Textures - can be set to the maximum
 2. Draw Distance - Set to the first step above the minimum
3. WaveSteward often uses the Teleport Stones, which are available in the normal and event shops.
4. And just to be clear, WaveSteward does not make the viewer crash. Often, when it crashes, it originates from settings in your PWI client.

Big thanks

Big thanks to the people sending in bug reports, wishes and tips.

Previous releases

New in this release 1.8-B - 25/05/2026

"Used the broom, but not as a witch."

Added

1. From now on, it is possible to collect the BH for DP(TT) 5-1, 5-2, and 5-3 in the dailies.
That is also the reason why the tab BH in MISC has been expanded, now showing all possible BH that can be mass collected, as well as the mass delivery of the BH.
2. **[In development]** Ability to kill the mobs and collect the drops during the dailies in Eastern Continent.
This function is still being developed, but has been released for manual testing.
If you want, you can test it out by using Info >> Debug. You will find a button [Monster Kill Test].

Currently, it contains both killing and collecting the drops.

The main points to address are buffing and the need for a stronger partner if you are too weak.

3. **[Extension]** -- Residual Lights on Eastern Continent - WaveSteward can now automatically collect 85 of the residual lights.

On the map below, the points displayed show where the lights are. The ones with a red circle cannot be picked up automatically and must be picked up manually, mostly through a quest.

IMPORTANT: You need to have the basic teleport points in the Eastern Continent opened.

Collecting the Residual Lights can be initiated using the button Residual Lights on the Tab Misc >> EC Farming.

You can find the above block dedicated to the Eastern Continent.

And if you want to see the map, just use the button on WaveSteward.

Please note that when WaveSteward is running quests, of course, the button will not react, as with all the buttons.

Modified

1. **[Bug]** -- One of the EC dailies was not closing correctly. The code has been updated so that he now correctly turns in the quest at the Holy Ritualmaster.
2. **[Bug]** -- One of the Eastern Continent dailies 1, where you had to go to a location near the Holy Ritualmaster, was failing with your character stuck in an infinite loop, as he could not reach his destination. This should be sorted out now.
3. **[Bug]** -- One of the Eastern Continent dailies 3, Monster Hunt (Mystic Cloudvale Archer) x 1, was failing as you dropped too close to the mobs and got killed. The landing point has been adapted to a more safer spot.
4. **[Bug]** -- Wrong application name in Windows Process Manager
5. **[GUI]** -- For visibility, some of the forms used in the GUI have the buttons reordered. This was done to reduce the proliferation of buttons in the wrong place and to give the user a better overview. The forms involved are:

Combat

Instead of a single form on WaveSteward, it has been split into three separate tabs: Mass Functions, Spam Functions and Main Char Functions.

Misc

The tab Farming contains all functions to access and, if possible, farm instances.

The Extra Farming tab contains all functions not related to instances, Northern Realm, or the Eastern Continent.

6. **[Experimental]** -- The experiment for reducing the drunkness in Trial 7 is still going on, but is not giving a correct result for the moment.

A very important remark -- To keep in mind. :

Due to changes in WaveSteward's Daily tab, please verify your settings there, as some of your dailies may have shifted. I tried to avoid it as much as possible.

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4. And just to be clear, WaveSteward does not make the viewer crash. Often, when it crashes, it originates from settings in your PWI client.

Big thanks

Big thanks to the people sending in bug reports and tips.

New in this release 1.8-A - 04/05/2026

"Only twelve hours to create a route. Could have been worse."

And some new stuff has happened in this release.

1. The Eastern Continent dailies have been reviewed and expanded. The highlighted numbers (in bold) are the extended ones.
 1. The count is currently:
 1. **30** Phase 1 quests
 2. 15 Phase 2 quests, of which some are very difficult shadow runs with jumping
 1. The good news is that one of the most difficult ones has been included.
If you recognise where this is, you will know which one it is.

But it stays a difficult one and is sadly prone to fail.

(Thanks to Mojra for spending hours to sort this one out. And I am sure that the catalogue of curses existing in the world has tripled three times in size since she started it.)
 3. **27** Phase 3 quests
 1. If you need to kill monsters, the bot will wait to continue till you have done it manually.
 2. You can find the name of the monster to kill and the quantity to kill in the log screen and in the notification window (A small aesthetic extension).
 3. WaveSteward will bring you as close as possible to the group of mobs you need to be near.
 4. And after the mob(s) are killed, you have 25 seconds to gather as much stuff as possible before continuing.
 5. Please note that some mobs need to be killed two times before the quest completes.
 4. **9** Phase 4 quests @ Holy Ritualmaster
 2. Sadly, I could not extend the Residual Light positions further. Working on it as much as possible. But on

the other side, you won't become lazy then, as you also need to do something.

3. There is now a hidden, sneaky ability in the WaveSteward that will probably help some people.

1. It concerns the delivery of artefacts. But I won't say more. Just play with it a bit, and you will see what it is. Anyway, you probably already heard about it through the grapevine and didn't dare speak up.

4. If you ever hit a daily quest on the Eastern Continent, which is still unknown, you can help us with it by collecting the route for it. On one of the following days, I will provide a document explaining how to do it.

Anyway, I think we have most of the quests already, but you never know that you are the lucky one to catch a new one.

5. And as usual, if you ran into an error, just screenshot it, send it over with a nice description on how you got there, what you were doing and so on, and we will try to look at it

6. Speed Dailies are starting differently and should open faster. Also, you have settings for this.

These new settings in Info >> Settings are available:

1. 1. Enable fast dailies starting (Experimental) - This is unchecked by default, so the system uses the original approach to open additional bots and start dailies on alts; when checked, the new approach is used, which opens a bit faster.
2. Speed Dailies start main character dailies also (Exp.) - Is by default Unchecked, but if you want the Main character to start his dailies also immediately after all alts have started, just check this box.

7. Some small changes:

1. Version number is now visible on the GUI of WaveSteward in the bottom right corner
2. An experimental function - never tested it out, so not sure if it works. On the tab Misc >> Special Event/etc. there is a button "Stop being drunk".

Not sure if this will even work in a Trial 7 after digging the apples, but worth a try.

If not working, it will disappear quickly.

But you can even Hotkey it for now.

A very important remark -- To keep in mind. :

Due to changes in the Daily tab of WaveSteward, please verify your settings there, as some of your dailies may have shifted. I tried to avoid it as much as possible.

Things in the pipeline

1. Adding DP quests to the Bounty Hunter group on the Daily tab
2. Solution for long-standing issues:
 1. [Speed Dailies] going in a loop and stopping to react
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 3. Failure to enter the homestead because it is not in a safe zone.
3. Ability to interrupt WaveSteward without requiring the user to close the application.

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1. Ability to predict the complete set of bosses and mobs in the WM instance. (We are getting there

slowly...)

2. Continue revamping the historically overloaded screens, which is still ongoing.
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Big thanks

Big thanks to the people sending in tips and routes like **Blenta**, **Mojra**, and others whose names I forgot to add here.

New in this release 1.7-B - 26/04/2026

*"Being in a hurry, always kicks you in the a**."*

So a new release, very quickly after the previous one in less than a week, but this is most of all to solve some issues that were in the last version. My excuses for the buggy last release.

1. A bug in opening the Eastern Continent quests has been resolved. Sorry for that.
2. Multi-sin life hunter spamming wasn't working correctly.

And some new stuff has happened in this release.

1. The Eastern Continent dailies have been reviewed and expanded.
 1. The count is currently:
 1. 26 Phase 1 quests
 2. 15 Phase 2 quests, of which some are very difficult shadow runs with jumping
 3. 17 Phase 3 quests
 1. Note: If you need to kill monsters, the bot will wait to continue till you have done it manually.
 2. You can find the name of the monster to kill and the quantity to kill in the log screen and in the notification window (A small aesthetic extension).
 4. 5 Phase 4 quests @ Holy Ritualmaster
 2. The collection of Residual lights is getting implemented. Currently, the first 30 are being collected. More to come.

You can find in below Misc >> Extra farming
3. A new way for WaveSteward to communicate with the user.

If there is some information that needs to be made over, the use of the standard Windows message box was a headache as it was usually hidden by the game, or even pausing the game.

This has now been replaced by a notification window that will appear on the bottom right of your screen, and looks like

Normally, they will close on their own, but you can close them more quickly by clicking the X in the top-right corner of the notification.
4. When opening additional WaveSteward using the buttons [SPEED BOTS] or [SPEED DAILIES], the splash screen is not displayed, but you will see in the status bar of the progress as well as in the LOG tab.

5. Some aesthetic redesign of the Daily tab has happened.

1. Some daily activities were by default exclusive in their actions, but could still be selected next to each other on the screen. This ability has not been removed.
2. In the Eastern Continent group, this has been extended a bit more.
The selection of your Pangu Soul Tier is now exclusive; you can only select one. But you can also check that you don't want to do a Pangu Soul tier at all, and this will disable all the other selections involved with it.

A very important remark:

Because of the changes in the Daily tab of WaveSteward, please verify your settings in there, as some shifts in your dailies may have happened. I tried to avoid it as much as possible.

Things in the pipeline

1. Adding DP quests to the Bounty Hunter group on the Daily tab
2. Solution for long-standing issues:
 1. [Speed Dailies] going in a loop and stopping to react
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1. Ability to predict the complete set of bosses and mobs in the WM instance. (We are getting there slowly...)
2. Continue with the revamping of the historically overloaded screens which is still on-going

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Big thanks to the people sending in tips and routes like **Blenta**, **Mojra**, and others whose name I forgot to add here.

New in this release 1.5-A

1. The collection of resources on Eastern Continent has been reviewed and updated:
 1. You will be collecting more resources now all the time. And now worries, you will even find resources that you can't see.
 2. The collection time has been reduced because the search path for resources has been reduced.
 3. Some flights have been remodeled, to try to limit the issues of being stuck against rocks or trees.
2. The alpha version for the Eastern Continent Dailies is from now on available within WaveSteward, but there are a number of points to keep in mind.
 1. Currently only level 1 is available. Other levels will follow at a later stage.
 2. Not all daily quests are available yet.
 1. If a quest is not yet available in WaveSteward, it will appear in the log of the bot, with the

quest identifier. You will need to do these quests then manually.

3. This is an alpha version, so please be aware that your quest will at moments fail, or your actor will be stuck during its quest. If this happens, please let us know, and describe the issue in detail so we can attend to it.

You will have sadly to restart WaveSteward to continue then. We are looking for a solution to be able to interrupt it without restart.

4. Currently the routes have been hard-coded, as work is being done on an update ability for the WaveSteward to learn new routes without needing new releases.
5. If you are interested to help, you can, as the bot will contain the draft functionality to record and store new routes which can be send over this channel or directly. Feel free to take contact for more information.

3. LifeHunter spamming has been extended for the assassins.

1. It is now possible when you have multiple assassins loaded on your computer, to have them all use at the same moment either LifeHunter spam, or Pangu LifeHunter spam if they have learned that skill already, on the same boss.

Please note that the alts do not need to be in a squad, just near to your main assassin character. And they will attack the boss if he is near enough.

So best way to use it, position your main assassin near target, call all your alts near with a mass move, and then execute the multi life hunter spam.

2. For this a new shortcut setting has been added.
3. This is currently in a beta version, so be prepared that sometimes it may fail, or you need to move your characters to make them stop.
4. The spamming uses for the moment the standard values of 100/10, and will be later extended to use the user defined values.

4. Phoenix Valley has been added to the dailies.

This will send your character to Phoenix Valley to collect experience and spirit.

1. It is currently just one character base, it is planned to extend this to an ability to include squad members running on the same computer.

2. Also the method how the killing is done still needs to be reviewed to optimize the effect.

5. To bring down the doubt if you have started WaveSteward or not, a small esthetic modification has been done to the starting of it.

A splash screen will open while starting it, showing information on the release, and also its progress in loading.

This is not finalised completely.

6. A review has been done on the LifeHunter and Pangu LifeHunter with user settings to work out some minor bugs in it.

7. Generic Pangu Spam has been temporary removed from WaveSteward to allow a more correct revamping and implementation.

8. WaveSteward is to remember its position and size, and not only its size in the next release, to bring down the need to move the window of WaveSteward on the screen to make it good visible.

This still needs to be finalized for the dialog boxes.

9. Ability to pause WaveSteward during Easter Continent daily runs to allow you to kill mobs as requested

10. Some esthetics and revamping of overloaded forms

Things in the pipeline

1. Ability to pause the bot during Easter Continent daily runs to allow you to kill mobs as requested
2. Solution for long-standing issues:
 1. [Speed Dailies] going in a loop and stopping to react
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1. Ability to predict the complete set of bosses and mobs in the WM instance. (We are getting there slowly...)
2. Continue with the revamping of the historically overloaded screens

Big thanks

Big thanks to the people sending in tips and routes like **Blenta**, **Mojra**, and others whose name I forgot to add here.

New in this release - 1.0

1. The LifeHunter ability has been revised following reports of a short freeze occurring at regular intervals, lasting about a bit less than a second.
This issue should no longer happen.
2. This version can collect resources on the Eastern Continent.
For that reason, a checkbox group has been added to the dailies list below the Northern Realm group.

Not all resources visible are being collected; this is not a malfunction but a decision not to overload the system. But when done on a daily basis, you will notice that the amount of resources in your Pangu bag grows steadily.

Please note that access to the Easter Continent with a 32-bit client is precarious.

For that reason, it is advised to lower your graphics settings to the minimum and ensure the quest list is no longer displayed on your screen.

DC or crash could still occur, but in a lesser amount, as it also depends on your computer running the client and the number of clients you have open at the same time.

This crashing is also the reason why the daily quests for the Eastern Continent and the Adventure Kingdom are kept at the bottom of the list, as both can cause your client to crash. By moving them to the back, it can be assured that the most important dailies have been finalised.

The route that is followed for the collection of resources is focused on the Northern side of the map, as normally there are no sealed caves that contain resources.

The current route is not ideal yet in terms of speed and flow, and will be revisited for optimization.

3. A new approach for the LifeHunter Spam functionality has been added, allowing you to configure the settings for the Cancel Delay and the Recast Delay to values of your choice.
This is at your own risk, as the default values of 100 milliseconds for Cancel Delay and 10 for Recast

Delay are ideal, though slightly less performant, but ensure your system does not lock up.

The settings can be done in [Misc >> Skill Settings].

The buttons save the values for each setting separately in the settings.ini file.

4. Two new skills with spamming ability have been added and are available in the [Combat] tab.

The PANGU LF Spam is identical to the standard LifeHunter spam for assassins and can also be configured in the section mentioned in point 2.

Next to that, there is also the ability now for clerics to spam using Cleric Cyclone, and again, you can personalise the speed in the form displayed in point 2.

Both spams can also be assigned to the same hotkey as the original one and are found below the hotkey fields for Sin: LF Spam.

Please note that both skills will need a GREEN glyph to be able to spam them.

5. An in-depth review of the WaveSteward tool's code has been performed. In addition to the above-mentioned additions, some minor issues and obsolete/unnecessary actions within the tool have been removed.